0.1 (1)

For a uniform disk of radius a, the surface mass density is

$$\sigma = \frac{m}{\pi a^2}.$$

The moment of inertia about the axis through its center of mass is

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^a \rho^2 \sigma \, \rho \, d\rho \, d\vartheta \\ &= 2\pi\sigma \frac{a^4}{4} = \frac{1}{2} m a^2. \end{aligned}$$

0.2 (2)

For the center of mass, the tangential equation of motion is

$$F - mg \sin\theta = m(R-a)\ddot{\theta}.$$

For rotation about the center of mass,

$$Fa = I\dot{\omega} = \frac{1}{2} m a^2 \dot{\omega}.$$

0.3 (3)

The no-slip condition requires the velocity of the contact point to vanish:

$$(R-a)\dot{\theta} + a\omega = 0.$$

0.4 (4)

Differentiating the no-slip condition gives

$$(R-a)\ddot{\theta} + a\dot{\omega} = 0.$$

Using the rotational equation of motion,

$$F = \frac{1}{2} m a \dot{\omega} = -\frac{1}{2} m (R-a) \ddot{\theta}.$$

Substituting this into the tangential equation,

$$\frac{3}{2} m (R-a) \ddot{\theta} = -mg \sin\theta,$$

so that

$$\boxed{F = \frac{1}{3} mg \sin\theta}.$$

0.5 (5)

The kinetic and potential energies of the disk (C) and the attached mass (D) are

$$\begin{aligned} T_C &= \frac{1}{2}m(R-a)^2\dot{\theta}^2 + \frac{1}{4}ma^2\omega^2, \\ T_D &= \frac{1}{2}mR^2\dot{\phi}^2, \\ V_C &= -mg(R-a)\cos\theta, \\ V_D &= -mgR\cos\phi. \end{aligned}$$

Thus, $T = T_C + T_D$ and $V = V_C + V_D$.

0.6 (6)

Since the semicircular track now rotates, the no-slip condition becomes

$$(R-a)\dot{\theta} + a\omega = R\dot{\phi}.$$

0.7 (7)

From the no-slip condition,

$$\omega = \frac{R\dot{\phi} - (R-a)\dot{\theta}}{a}.$$

The Lagrangian is then

$$\begin{aligned} L &= T - V \\ &= \frac{1}{2}m(R-a)^2\dot{\theta}^2 + \frac{1}{4}m\big[(R-a)\dot{\theta} - R\dot{\phi}\big]^2 + \frac{1}{2}mR^2\dot{\phi}^2 \\ &\quad + mg(R-a)\cos\theta + mgR\cos\phi \\ &= \frac{3}{4}m(R-a)^2\dot{\theta}^2 - \frac{1}{2}m(R-a)R\dot{\theta}\dot{\phi} + \frac{3}{4}mR^2\dot{\phi}^2 \\ &\quad + mg(R-a)\cos\theta + mgR\cos\phi. \end{aligned}$$

Applying the Euler–Lagrange equations to θ and ϕ, respectively, gives

$$\begin{aligned} -mg(R-a)\sin\theta &= \frac{3}{2}m(R-a)^2\ddot{\theta} - \frac{1}{2}m(R-a)R\ddot{\phi}, \\ -mgR\sin\phi &= \frac{3}{2}mR^2\ddot{\phi} - \frac{1}{2}m(R-a)R\ddot{\theta}. \end{aligned}$$

0.8 (8)

For small oscillations, using $R - a \approx R$, $\sin\theta \approx \theta$, and $\sin\phi \approx \phi$, the equations become

$$\begin{aligned} -g\theta &= \frac{3}{2}R\ddot{\theta} - \frac{1}{2}R\ddot{\phi}, \\ -g\phi &= \frac{3}{2}R\ddot{\phi} - \frac{1}{2}R\ddot{\theta}. \end{aligned}$$

Assuming a linear combination

$$\psi_\lambda = \theta + \lambda\phi, \qquad \lambda = \pm 1,$$

we obtain

$$-g\psi_\lambda = R\left(\frac{3}{2} - \frac{\lambda}{2}\right)\ddot{\psi}_\lambda.$$

For $\lambda = +1$,

$$\ddot{\psi}_+ + \frac{g}{R}\psi_+ = 0, \qquad \psi_+ = \theta + \phi.$$

For $\lambda = -1$,

$$\ddot{\psi}_- + \frac{g}{2R}\psi_- = 0, \qquad \psi_- = \theta - \phi.$$

Therefore, the two normal-mode angular frequencies are

$$\boxed{\Omega_+ = \sqrt{\frac{g}{R}}, \qquad \Omega_- = \sqrt{\frac{g}{2R}}}.$$